Proof by Induction: General

Questions 1–15

Question 1.

a) Prove by induction, for all integers $n \geq 1$, that

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

b) By considering $\frac{1}{(\cos\theta + i\sin\theta)^n}$ extend the proof to negative integers.

Question 2. Prove by induction, for all integers $n \leq -1$, that

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

Question 3. Prove by induction for all integers $n \geq 1$ that

$$\frac{1}{3} \times \frac{3}{5} \times \frac{5}{7} \times \ldots \times \frac{2n-1}{2n+1} = \frac{1}{2n+1}$$

Question 4. Prove, using mathematical induction, that

$$\sum_{r=0}^{n} \binom{n}{r} = 2^n$$

for all integers $n \geq 0$

Question 5. Use proof by induction to show that

$$\sum_{r=1}^{n} \ln(2r-1) = \ln\left(\frac{(2n)!}{2^n n!}\right)$$

for all positive integers $n \geq 1$

Question 6. The function $\mathrm{f}(x)$, for $x \neq -1$ is defined by

$$\mathrm{f}(x) = \frac{x}{x+1}$$

Prove that for all positive integers $n \geq 1$,

$$\mathrm{f}^n(x) = \frac{x}{nx+1}$$

holds true, where $\mathrm{f}^n(x)$ is the nth composition of $\mathrm{f}(x)$

Question 7. Use proof by induction to prove that

$$\frac{d^n}{dx^n}(x^n) = n!$$

for all positive integers $n \geq 1$

Prerequisite Knowledge for Q8: Differentiation of Standard Functions and Chain Rule

Question 8. Prove by induction, for all integers $n \geq 1$ where a is a positive constant, that:

a) $\frac{d^n}{d\theta^n}(\sin a\theta) = a^n \sin\left(a\theta + \frac{n\pi}{2}\right)$

b) $\frac{d^n}{dx^n}(\ln(1+ax)) = \frac{(-1)^{n+1}(n-1)!a^n}{(1+ax)^n}$

Prerequisite Knowledge for Q9: Maclaurin Series

Question 9. It is given that $y = e^{ax+b}$ where a and b are constants and $a \neq 0$

a) Prove by mathematical induction, for all integers $n \geq 1$, that

$$\frac{d^n}{dx^n}(e^{ax+b}) = a^n e^{ax+b}$$

b) Hence find the first three terms of the Maclaurin series for e^{ax+b}, simplifying your answer as far as possible. You are not required to show that this series converges.

Prerequisite Knowledge for Q10: Product Rule

Question 10. Use proof by induction to prove that

$$\frac{d^n}{dx^n}(e^x \cos(\sqrt{3}x)) = 2^n e^x \cos\left(\sqrt{3}x + \frac{\pi}{3}n\right)$$

for all positive integers $n \geq 1$

Prerequisite Knowledge for Q11-Q13: Differentiation of Hyperbolic Functions

Question 11. Given that $y = \sinh x$, prove by induction that

$$\frac{d^n y}{dx^n} = \sinh x$$

for all even positive integers $n \geq 2$

Prerequisite Knowledge for Q12-Q13: Product Rule

Question 12. Given that $y = x \sinh x$, prove by induction that

$$\frac{d^{2n-1} y}{dx^{2n-1}} = x \cosh x + (2n-1) \sinh x$$

for all positive integers $n \geq 1$

Question 13. Given that $y = e^x \cosh x$, prove by induction, for all positive integers $n \geq 1$, that

$$\frac{d^n y}{dx^n} = 2^{n-1} e^x (\cosh x + \sinh x)$$

Question 14. Given that

$$f_n = x - x^2 + x^3 - \cdots + (-1)^{n-1} x^n$$

for $x \neq 1$, prove by induction, for all integers $n \geq 1$, that $f_n(x) = \frac{x(1-(-x)^n)}{1+x}$

Question 15. Prove by mathematical induction that

$$(1+x)^n \equiv 1 + {}^nC_1x + {}^nC_2x^2 + \cdots + {}^nC_nx^n$$

for all positive integers $n \geq 1$